GOVARTHAN K S

Electronics & Communication Engineering Student | Embedded Systems & Electronics

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PROFESSIONAL SUMMARY

Second-year Electronics and Communication Engineering student with hands-on experience developing ESP32-based systems and prototypes spanning electrical fault monitoring, smart irrigation, AI-enabled agriculture and railway applications. Interested in embedded systems, electronics and semiconductor technology, with foundational programming skills and experience presenting technical topics.

EDUCATION

V.S.B Engineering College — B.E. Electronics and Communication Engineering

Currently pursuing: Second Year | Expected graduation: 2029

TECHNICAL SKILLS

Programming: C, C++, Java, Python

Embedded systems: ESP32, Embedded C, Arduino IDE, microcontroller interfacing

Electronics: Current and voltage sensing, soil-moisture sensing, relay modules, DC pump interfacing, power-supply and buck-converter concepts

Web: HTML

PROJECTS

COMPLETED PROJECT

1. Smart Water Irrigation System using ESP32

- Built an automated irrigation prototype using ESP32, a soil-moisture sensor, a 5V relay module and a 5V DC water pump.
- Programmed moisture-based pump control using Arduino IDE to automate watering based on sensor readings.

DEVELOPED PROTOTYPES

2. VOLT SENTINEL – Smart Wire Fault Detection System

- Developed a prototype concept for monitoring low-voltage AC overhead distribution conductors using ESP32 with current and voltage sensing.
- Worked on identifying abnormal electrical conditions and presenting monitoring information through a dashboard.

3. Industrial Wire and Fault Detection System

- Developed an ESP32-based electrical monitoring prototype integrating current and voltage sensors.
- Worked with a buck converter and SMPS power supply, alongside a web dashboard concept for displaying system data.

4. AI-Powered Crop Health Monitoring System

- Developed a prototype solution combining agricultural sensing and AI-based analysis for crop-health monitoring.
- Explored multispectral/hyperspectral imaging and soil parameters such as moisture, pH and temperature to assess crop stress and potential risks.

5. AI Crop Yield Prediction and Optimization

- Developed a prototype concept for estimating crop yield from agricultural data and exploring optimization recommendations.
- Worked on a web interface using HTML and Bootstrap, with a prediction-API integration concept.

6. AI-Based Railway Track Fitting Identification

- Developed a prototype concept for railway track-fitting identification using laser-based identification and QR-marking approaches.
- Focused on supporting component identification and railway inspection workflows.

7. Automated High-Current Short-Circuit Test System for MCB Compliance

- Developed a prototype for exploring automation in MCB short-circuit testing, with reference to IEC 60898-1:2015.
- Worked on electronic control and test-process automation concepts for more consistent testing.

INTERNSHIPS & CERTIFICATIONS

Semiconductor Manufacturing Internship — Lakshmi Automations

Pega Internship — Internship completed; specific role and tasks not provided

Fundamentals of Semiconductor Devices — GUVI

ACHIEVEMENTS & TECHNICAL ACTIVITIES

- Participated in two hackathon opportunities and progressed to the screening stage.
- Presented semiconductor-related technical papers three times.
- Participated in a semiconductor workshop.

CAREER INTERESTS

Embedded Systems Development • Electronics • Semiconductor Technology • Microcontrollers • Hardware–Software Integration